
CAPSTONE PROJECT

AI FASHION DESIGN GENERATOR

Presented By

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INTRODUCTION & BACKGROUND



01 / INSIGHT

Manual Ideation

Creating and iterating on fashion concepts manually requires significant time and expertise.



02 / BARRIER

Specialist Tools

Professional design software can create technical and accessibility barriers for non-expert users.



03 / INEFFICIENCY

Fragmented Flow

Design generation, product discovery, and virtual try-on are commonly handled as separate processes.



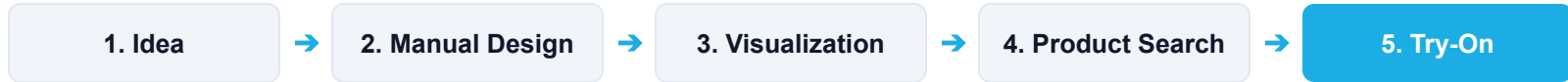
04 / SOLUTION

Opportunity

Generative AI can combine these stages into one unified fashion workflow.

PROBLEM STATEMENT

Traditional Workflow



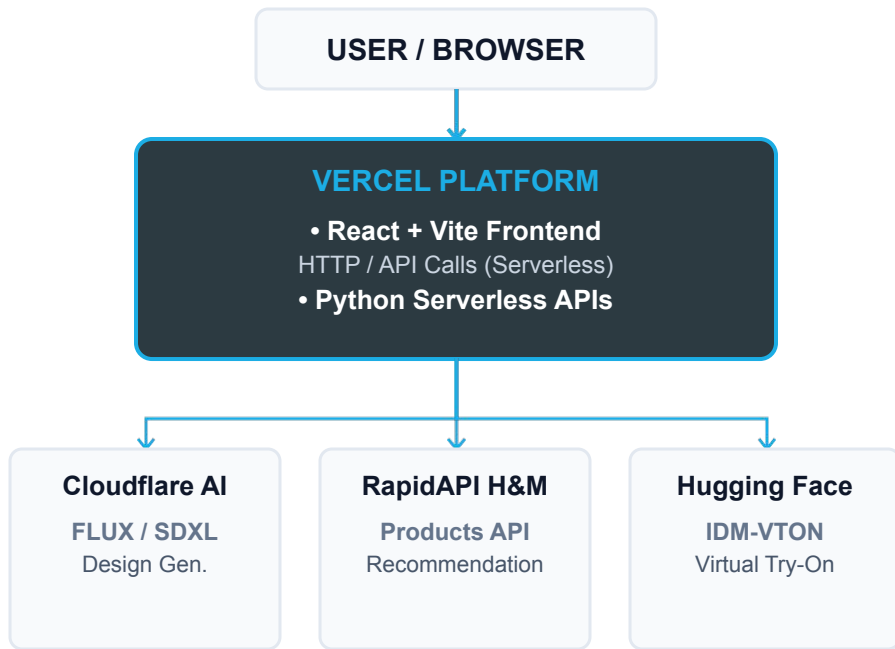
Traditional Barriers

- **Lack of Technical Skills:** Many students want to explore fashion design but lack the necessary artistic or technical training [cite: 4].
- **High Entry Barriers:** Traditional prototyping and design tools are expensive, slow, and resource-intensive [cite: 4].

The AI Opportunity

- **Text-to-Design:** Instantly synthesize unique, customizable clothing designs directly from simple text prompts [cite: 4].
- **Intelligent Matching:** Suggest similar, highly affordable products online to connect digital designs with physical retail [cite: 4].
- **Enhanced Accessibility:** Empowers aspiring creators by making fashion design deeply interactive, accessible, and fun [cite: 4].

SYSTEM ARCHITECTURE



API Routes

`/api/design` Cloudflare Workers AI
`/api/products/search` RapidAPI H&M Store
`/api/try-on` Hugging Face IDM-VTON
`/api/health` Service health check

Key Architecture Benefit

Single Vercel deployment handles both the React frontend and Python serverless API layer, while external AI services are securely accessed through server-side APIs.

Secrets remain server-side • Serverless • Scalable • No separate backend deployment

KEY FEATURES / MODULES



AI Fashion Studio

Create fashion designs from natural-language prompts.



Fashion Spec

Extract structured attributes like fabric, fit, and budget.



Image Generation

Generate fashion concepts using multiple AI models.



Product Match

Find matching real products through the H&M API.



Virtual Try-On

Generate interactive AI try-on results using IDM-VTON.



Collections

Save and manage your generated fashion designs.



Tech Pack

Generate structured technical fashion specifications.

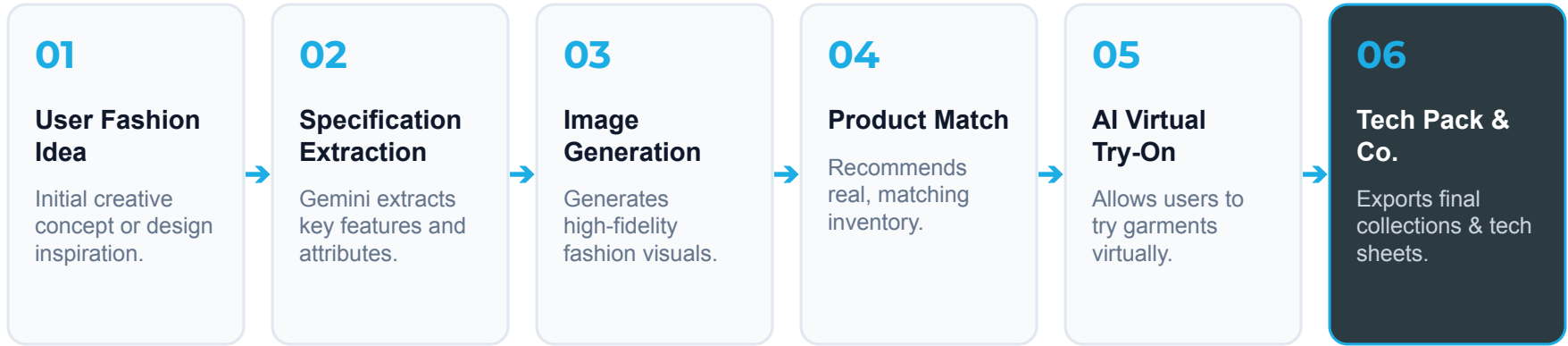


Utilities

Download designs and copy specifications instantly.

PROPOSED SOLUTION

Unified AI Fashion Workflow



A unified end-to-end fashion ideation and visualization platform powered by Generative AI.

System Development — Technology Used



Frontend

- React 19
- Vite 8
- Tailwind CSS 4
- Lucide React



Backend / API

- Python
- Vercel Serverless
- Pydantic
- HTTPX



Artificial Intelligence

- Gemini 2.5 Flash
- Cloudflare AI
- IDM-VTON



External Services

- Hugging Face
- H&M Store API
- Vercel

Note: Your technology choices should match the latest repository rather than the older FastAPI architecture shown in the sample.

ALGORITHM & DEPLOYMENT

A. Recommendation Algorithm

Metadata Matching Weights:

Category Match 40%

Color Match 25%

Budget Fit 20%

Style / Brand 15%

Scoring Formula:

Score = Cat + Color + Budget + Style

** Structured metadata matching only; does not use vector embeddings for this ranking.*

B. Deployment & Architecture

CI/CD & Hosting Pipeline:

GitHub → Vercel → React + Python APIs → External APIs

Production API Routes:

GET /api/design

POST /api/products/search

POST /api/try-on

GET /api/health

RESULT

Project Outcomes

- ✓ AI-based fashion concept generation
- ✓ Structured fashion specification extraction
- ✓ Multiple AI image generation models
- ✓ Real H&M product recommendations
- ✓ IDM-VTON virtual try-on integration
- ✓ Before / After try-on experience
- ✓ Collections and Tech Pack functionality
- ✓ Download and copy utilities
- ✓ Serverless Vercel architecture
- ✓ Security and input validation



Key Result

Successfully integrated multiple AI and external services into a unified fashion design workflow.

CONCLUSION

OVERVIEW

Unified Platform

The **AI Fashion Design Generator** demonstrates how Generative AI, product APIs and computer-vision-based virtual try-on can be combined into a unified fashion platform.

KEY IMPACT

The project reduces fragmentation in the fashion ideation workflow by bringing design generation, product discovery and virtual try-on into one application.

ARCHITECTURE

Connected Workflow

The system seamlessly connects each stage of fashion creation:

01 IDEATION

02 GENERATION

03 RECOMMENDATION

04 TRY-ON

FUTURE SCOPE

AI & Virtual Try-On

- Improved garment extraction
- Better garment preprocessing
- Higher-quality virtual try-on
- More advanced fashion generation

Recommendation

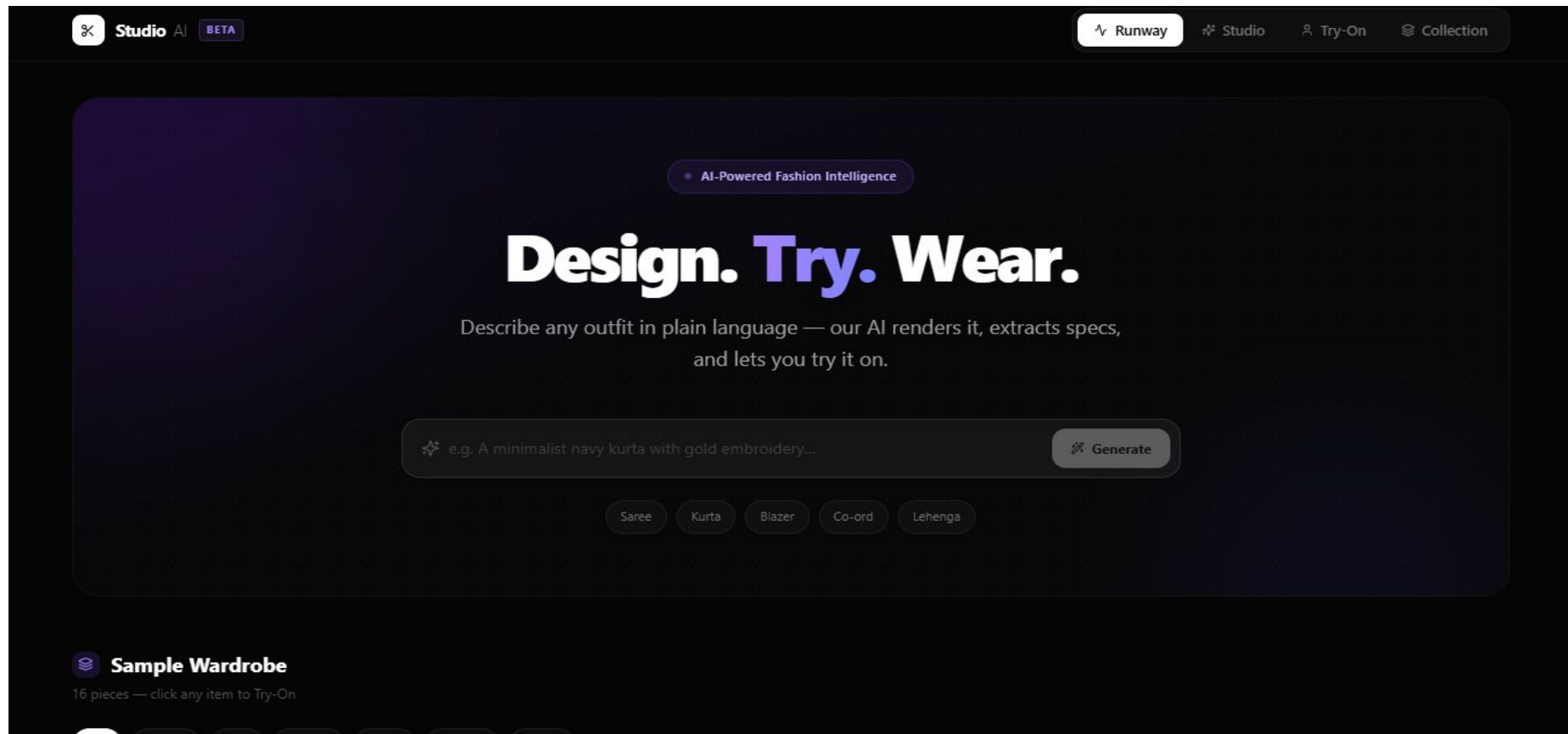
- More fashion marketplaces
- Personalized recommendations
- Advanced product similarity

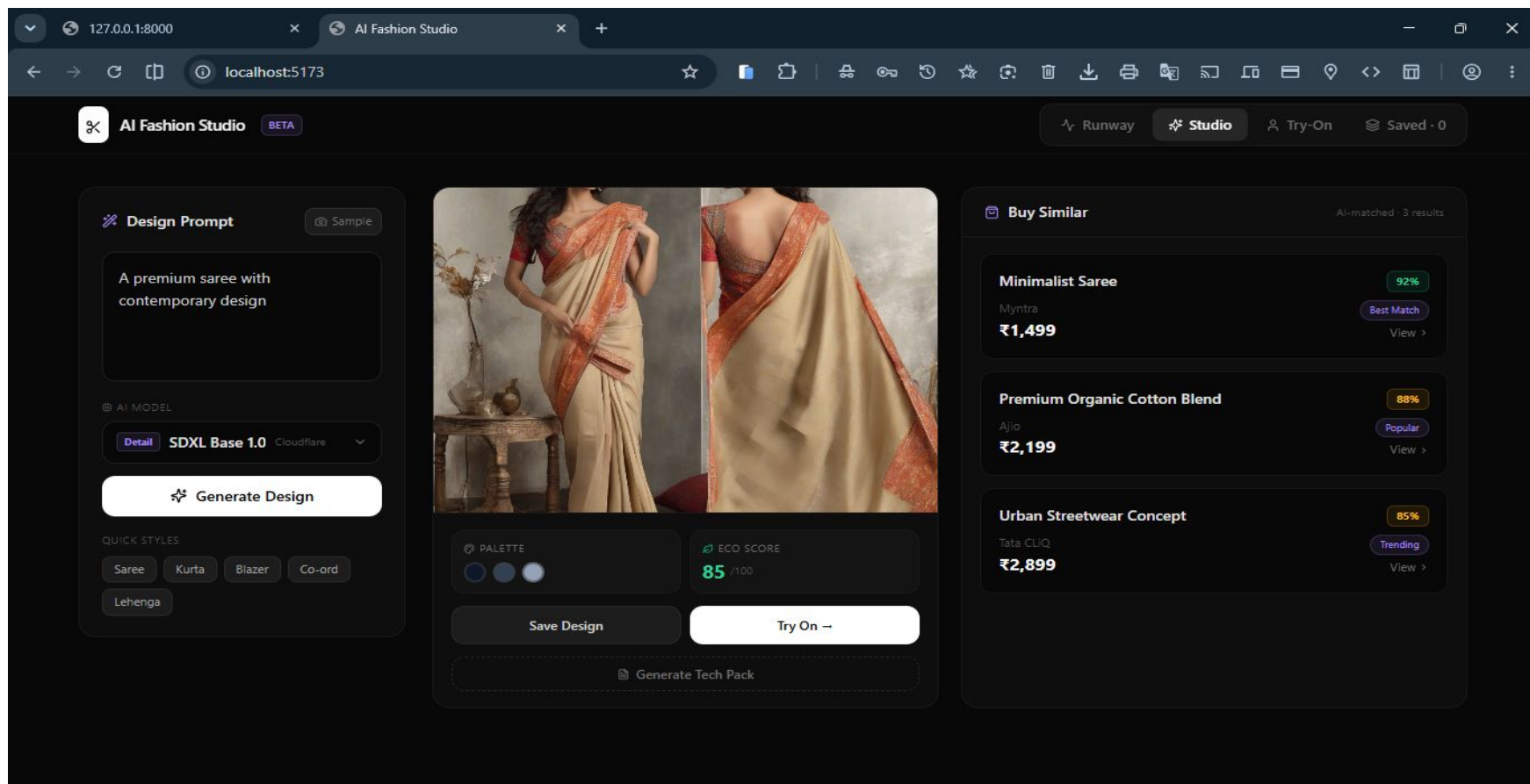
Platform

- User authentication
- Cloud-based collections
- Analytics dashboard
- Mobile application

Please Note: These items represent future roadmap scope and are not currently completed features.

RESULTS & SCREENSHOTS





Virtual Try-On

Upload your photo, pick a garment, then process

1 YOUR PHOTO



2 GARMENT



3 RESULT

AI is processing — this may take ~30–60 s...



Studio AI

BETA

Runway

Studio

Try-On

Collection

All (70)

Female (50)

Male (20)



Elegant Formal Look



Street Style Casual



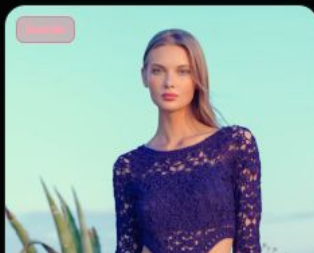
Summer Floral Dress



Minimal White Outfit



Boho Chic Look



REFERENCES

Project & Code

AI Fashion Design Generator

IBM Internship 2026

- GitHub Repository:

[AI-Fashion-Design-Generator-IBM-INTERSHIP-2026](#)

Third-Party Model

- IDM-VTON

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Technologies & APIs

- Google Gemini
- Cloudflare Workers AI
- RapidAPI H&M Store API
- Hugging Face
- IDM-VTON
- Vercel

Documentation

- React
- Vite
- Tailwind CSS
- FastAPI & Python
- Vercel Platform Docs

References & Standards: Keep this project structure clean, modular, and fully documented.

DEPLOYMENT & REPOSITORY



Live Demo

Deployment Platform:

Vercel Platform

Verified URL:

<https://ai-fashion-design-generator-ibm-int.vercel.app/>



GitHub Repository

Source Code & Documentation:

github.com/Piyu242005/AI-Fashion-Design-Generator-IBM-INTERSHIP-2026



Technologies

- React
- Vite
- Python
- Vercel

References & Standards: Keep this project structure clean, modular, and fully documented.

THANK YOU

"From idea to design to try-on — powered by AI."

Piyush Ramteke

IBM Internship 2026

AI Fashion Design Generator

